



SCI_LAUNCH_A2_W07 · LAUNCH Science

Autumn 2 Week 7 teacher note

Objective (verbatim): Topics 2 to 3 assessment (Dec).

Period sequence: This is the week's L1 lesson. L2 is lesson library SCI_L_A2_W7L2; L3 is lesson library SCI_L_A2_W7L3, the existing 24-mark, 24-minute test with a separate answer-free pupil paper and mark scheme. Do not set a second test.

Scheme of work: LAUNCH scheme of work, Autumn 2 week 7.

Pupil words: I am learning to use cell and inheritance models to show what I know and choose one next step.

Reading gate: LAUNCH 11+; rough FK grade 6.5, age proxy ≈ 12.5 , 1000 pupil-visible words excluding marked adult guidance. Formula is a rough screen, not an individual reading assessment.

Pearson foundation route: GCSE Biology 1BI0 Paper 1 and Combined Science 1SC0 Biology Paper 1, shared points 2.1–2.3; 3.5, 3.12–3.14 and 3.16 (actual shared Foundation content assessed). Official specification links: <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-biology-spec.pdf> and <https://qualifications.pearson.com/content/dam/pdf/GCSE/Science/2016/Specification/gcse-combinedscience-spec.pdf>. The workbook's teaching approach is not a new specification point. No Biology-only B/HT point is assumed for Combined Foundation.

Gatsby 4: A biomedical science technician records evidence from models and clearly separates observation from inference.

Sequence, success, retrieval

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40 minutes. Three one-minute resets are inside We do, Independent and Review; Arrival and resets make about eight minutes for settling and transitions.

Success at Starter, checked at Exit: I can order copy, separate and split for mitosis.; I can mark one gene as part of DNA.; I can use a simple cross as probability, not a promise.

Retrieval from SoW: "last term" uses the Autumn 1 block in the supplied 2026/27 workbook. If a literal previous academic term is required, staff must check actual teaching rather than invent it.

When	Question	Answer	Exact SoW cell
last lesson	Which named source supports our cystic fibrosis card?	The NHS page and NHS Genomics Education Programme.	LAUNCH scheme of work, Autumn 2 week 6
2-4 weeks	How does a gene relate to DNA?	A gene is a section of DNA.	LAUNCH scheme of work, Autumn 2 week 4
last term	Which part of an animal cell holds chromosomes?	The nucleus.	LAUNCH scheme of work, Autumn 1 week 1

Word meanings and examples (same in lesson and organiser):

Word	Tier	Definition	Example sentence
mitosis	3	division of the nucleus to separate matching chromosome sets	Mitosis helps make matching body cells.



Word	Tier	Definition	Example sentence
gene	3	a section of DNA that codes for a protein	A gene is a part of the DNA string.
allele	3	a version of a gene	A and a stand for two alleles in our invented model.
evidence	2	a fact or model detail that supports a claim	The labelled chromosome sets are evidence for my answer.

Faded example: I do full: Parallel example (not one of the check items): label a body-cell model with copied chromosome sets, then explain that mitosis separates them into two nuclei. Point to a model detail that supports the answer. Now cover the labels for the independent check. We do part-done: Parallel example: the chromosome sets are copied. Mitosis moves the sets to _ nuclei; the evidence detail is _. Removed step: State that matching chromosome sets go into two nuclei and cite a labelled detail. Independent uses no filled answer. Fade next: At the following review, remove the example's sentence stem and invite an independent evidence explanation.

Post-I-do show-me: Show me: does "describe" need a model detail or a personal story? Expected: A scientific model detail. Every pupil shows a board, point or card. Re-model the named step if insecure.

F/M/H and named stretch

We do F support: Use command words and a different model first; put it aside before independent items.

We do M: Explain which one model detail supports a claim, without using a check item.

We do H stretch: Give a limit of the rehearsal model: a diagram shows order but not every molecular step.

Independent F: In the independent check, order the three stages and mark one gene on the model. Stem: First ...; then ...; last ...

Independent M: Explain matching body cells and why a gene is only part of DNA. Stem: The model shows ... because ...

Independent H: Use $Aa \times aa$ to reject the claim that a probability promises an exact family count. Stem: The model chance is ...; the real count may differ because ...

Go deeper: Which model feature supports an explanation that transfers to a fresh example, and which detail is only the model?

Named early-finisher stretch — Error classifier after check: After submitting the check, compare two wrong cross explanations. Name each misconception and a precise repair without altering the checked answer. This is a harder correction/evaluation, not a repeated item.

On-screen interactions: Before the independent check, match model claims to their evidence. uses native selects with a check/reset and aria-live feedback; We do 2 is a native 4-option all-pupil hinge without auto-grading. No storage. Original labelled diagram is simplified, not to scale. Its labels are checked in the fact ledger.

SoW practical status: no core practical is claimed. Prepared model pieces are an optional handling rehearsal with the pupil-facing Risk/Safe lines below.

AfL hinge and misconceptions

Question at We do 2: Which paired statement is scientifically accurate?



Option	Correct?	Diagnosis
A. Mitosis halves chromosome number; a gene is the whole genome.	no	Confuses mitosis with meiosis and gene with genome.
B. Mitosis removes DNA; an allele is a separate cell.	no	Treats DNA as used up and allele as a cell.
C. Mitosis makes one unsplit cell; DNA is only in proteins.	no	Misses cell splitting and confuses DNA with protein.
D. Mitosis gives matching body cells; a gene is a section of DNA.	yes	Secure cell-cycle and gene distinction.

Decision: About 3 in 4 or more right → begin the check. Otherwise re-model a two-colour copy/separate/split strip and a short gene mark on a DNA string, then ask the parallel hinge. Do not pre-answer the three check items. Parallel hinge: What shows a gene on our string model? A. A marked section of the DNA string (expected). B. The whole organism. C. A cell outside DNA. D. An empty grid square.

Misconception	How to spot	Correction
Mitosis halves chromosomes.	Pupil chooses A, the halving claim.	Use a separate two-colour strip to show copying and matching sets; do not coach the assessment response.
A gene is the whole genome.	Pupil chooses A, calling a gene the whole genome.	Show a short mark on a separate DNA string, then leave the check item unsupported.

Planned adaptations

Broad area	EEF five-a-day	Concrete action
Cognition & Learning	scaffolding	Keep the cell and allele model in view; remove one cue for We do, then ask pupils to repeat a fresh example without it.
Communication & Interaction	flexible grouping	Offer the key word card and a private partner turn before speaking to the group; accept point, sign or AAC.
SEMH	cognitive/metacognitive strategies	Check in before the top question and hinge; offer a private first answer, agreed pause and one-step return.
Sensory/Physical	technology	Provide an enlarged high-contrast model and keyboard focus; allow speech-to-text or exact scribing.
EAL (language, not SEN)	explicit instruction	Pre-teach defined science words with examples; let pupils plan in any language while keeping M/H reasoning open.

EAL language support remains available at M and H; the challenge stays open. Speak before write; the organiser's "My language" column is blank, without machine translation. Accept point, sign, speak, draw, type, write, exact scribing and staff-captured photo. Offer speech-to-text, TTS and immersive reader.

TA Focus and prompt ladder



At Independent choose at lesson start: teacher works with supported pupils and TA holds group, or TA supports and teacher holds group. Swap across the week. No public cold-calling is required; this is an invitation bank.

I do: "Watch how I use a different model detail in my answer."

We do: "Which detail can you point to as evidence?"

Independent: "Read the command. Choose one valid response mode; this item is yours."

Least prompt first; no physical prompting: Wait five seconds for a self-start with command words. → Point to the command word, not the answer location. → Ask "Which item are you on?" without hinting at content. → Model a parallel non-test item only; pupil then returns to a fresh check item. → Correct the procedure (for example, which box to use), never a test answer.

Exit handoff: adult states deepest prompt level, valid response mode and what pupil did independently; other adult chooses the next prompt.

Route	Invitation	Expected answer
F	Which step copies DNA before mitosis?	Interphase.
F	Is a gene part of DNA or a whole cell?	Part of DNA.
M	Why can two body cells match after division?	Copied chromosome sets separated into two new cells.
M	What is an allele?	A version of a gene.
H	In $Aa \times aa$, what does two aa squares out of four mean?	A model chance of one in two, not an exact family count.
H	Which diagram detail would you cite for a gene claim?	The marked DNA section, not the full chromosome.

Feedback, EHCP broad areas and regulation

Codes exactly: VF · WS · I · NS(+) · E · R · // · ?. VF at We do or Review; WS when supported with recorded prompt level; I when independent at Independent/Exit; NS(+) precise next step at Exit; // a real self-edit at Review; ? adult reads the changed words back on request; R after the pupil responds to NS(+) next lesson and staff judge, never automatic; E only for an actual curriculum observation.

What I record and where: note hinge response split and the named misconception on the staff quick-record grid; record actual code and prompt level in feedback. E: evidence captured on the school's evidence app. Never gather evidence during distress or store pupil data in HTML.

Pathways: high support, VF/WS regular. Build: WS/I mix. Grow: mostly independent, self-edit and response. Launch: independent self-assessment.

Regulation: Check in before the hinge, top rung and Independent. At the agreed pause signal give space and a single return step. Collect no evidence while a pupil is distressed. Re-enter at the last completed step, with one small choice and private correction.

lesson record poster: LAUNCH_SCI_A2_W07. Staff device or projector only; pupil phones not required.

Lundy Exit: pupil can say, point, sign, draw or write; teacher hears the answer and changes the next model/prompt. Four answers: goal at top, actual Review feedback, Exit NS(+), success criteria at Starter and Exit.

Sources, checks and limits



Every scientific and diagram label claim is listed in LESSON_SOURCE.json with a named official source and status. False distractors are deliberately incorrect and diagnosed. Read source claims in context; no medical claim about an individual. The model match is a rehearsal on different prompts; keep it away before the independent check. No auto-marking and no saved score. “Dec” is the workbook label, not a newly invented assessment date.

Prepared model handling · practical safety

Kit: Large prepared chromosome strips and allele tiles, cell and cross mats, tray

Method: Use the prepared model pieces to rehearse each check, then record individual answers.

Pupil Risk: Loose model pieces could fall or be mouthed.

Pupil Safe: Use large pieces on a tray. Keep them out of mouths and collect them after.

Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

SLT planning and evaluation

Lesson code: SCI-LAUNCH-A2-W07

SoW reference: LAUNCH scheme of work, Autumn 2 week 7

Exact SoW outcome: Topics 2 to 3 assessment (Dec).

Evidence (scheme of work): Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Evidence in this lesson: Edexcel 1BI0 Paper 1 (1F) topic tests; AQA UAS science units. F: topic quizzes, practical write-ups. S: Topics 1 to 3 assessment (Dec).

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-LAUNCH-A2-W07. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Look at this week's familiar cell and allele models. Point to one detail you notice.

Staff: Offer a private, predictable start. Do not require previous knowledge.

Starter (3 min): Recall three ideas from earlier lessons. Show one answer at a time. Staff: All pupils respond; reveal the answers only after attempts. Revisit success criteria.

I Do (5 min): Watch the full model. Point to the step that answers today's question. Staff: Model one complete example; check every pupil's response to the show-me question.

We Do (7 min): Complete the missing step in the part-done model. Speak, then match the cards. Staff: Wait five seconds, gesture, then cue if needed. Support the route without supplying the missing answer.

We Do 2 (5 min): Choose one hinge answer. Show it at the same time as the group. Staff: Collect simultaneous responses, apply the decision rule and parallel hinge in the staff note.

Independent (8 min): Complete a short three-part check: order copy/separate/split, mark a gene on DNA, and use Aa x aa to state a possible pair and chance. Staff: At lesson start choose one adult with supported pupils and one holding the group; swap during the week. Record independence and prompt level.

Review (4 min): Tell a partner what you changed after feedback. Mark a real edit // and read it back when asked ?. Staff: Give a precise VF or NS(+) only when it actually happens. Invite a true self-edit and read-back.



Exit (3 min): Show what you know with a model and explain one reason. Staff: Hear the answer privately in a valid mode and use it to change the next example or prompt.

Planned adaptation — cognition learning: Keep the cell and allele model in view; remove one cue for We do, then ask pupils to repeat a fresh example without it.

Planned adaptation — communication interaction: Offer the key word card and a private partner turn before speaking to the group; accept point, sign or AAC.

Planned adaptation — semh: Check in before the top question and hinge; offer a private first answer, agreed pause and one-step return.

Planned adaptation — sensory physical: Provide an enlarged high-contrast model and keyboard focus; allow speech-to-text or exact scribing.

Planned adaptation — eal: Pre-teach defined science words with examples; let pupils plan in any language while keeping M/H reasoning open.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: flexible grouping; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Complete a short three-part check: order copy/separate/split, mark a gene on DNA, and use Aa × aa to state a possible pair and chance. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Spring 1 W01. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Spring 1 W01: Explain variation & evolution by natural selection.

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: State that matching chromosome sets go into two nuclei and cite a labelled detail. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "Which paired statement is scientifically accurate?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Look at this week's familiar cell and allele models. Point to one detail you notice. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Complete a short three-part check: order copy/separate/split, mark a gene on DNA, and use Aa × aa to state a possible pair and chance. Offer the agreed accessible



input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use mitosis, gene, allele, evidence with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.